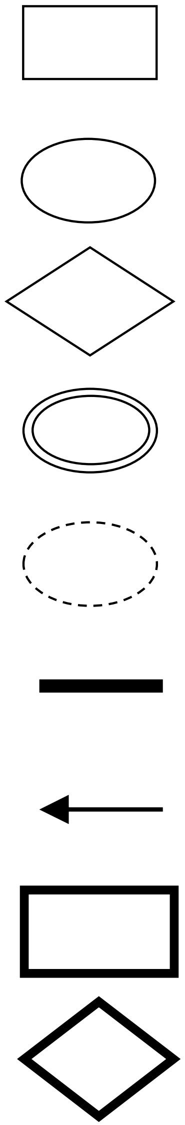
CS240: Tutorial 01

Review on E-R Diagram



- Rectangle – entity set
- Ellipse – attribute
- Diamond – relationship set
- Double ellipse – multi-valued attribute
- Dashed ellipse – derived attribute
- Thick line – total participation
- Arrow – key constraint
- Thick rectangle – weak entity set
- Thick diamond – relationship set for weak entity set

Exercise 1

- A university registrar's office maintains data about the following entities:
  a) **course**, including number, title, credits, syllabus, and prerequisites;
  b) **course offering**, including course number, year, semester, section number, instructor(s), timings, and classroom; (course offering depends on the course that university provided)
  c) **student**, including student-id, name, and program;
  d) **instructor**, including identification number, name, department, and title.
  - Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled.
- Construct an E-R diagram for the registrar's office. Document all assumptions that you make about the cardinality constraints and participation constraints.

Entity sets

"A university registrar's office maintains data about the following entity sets:"

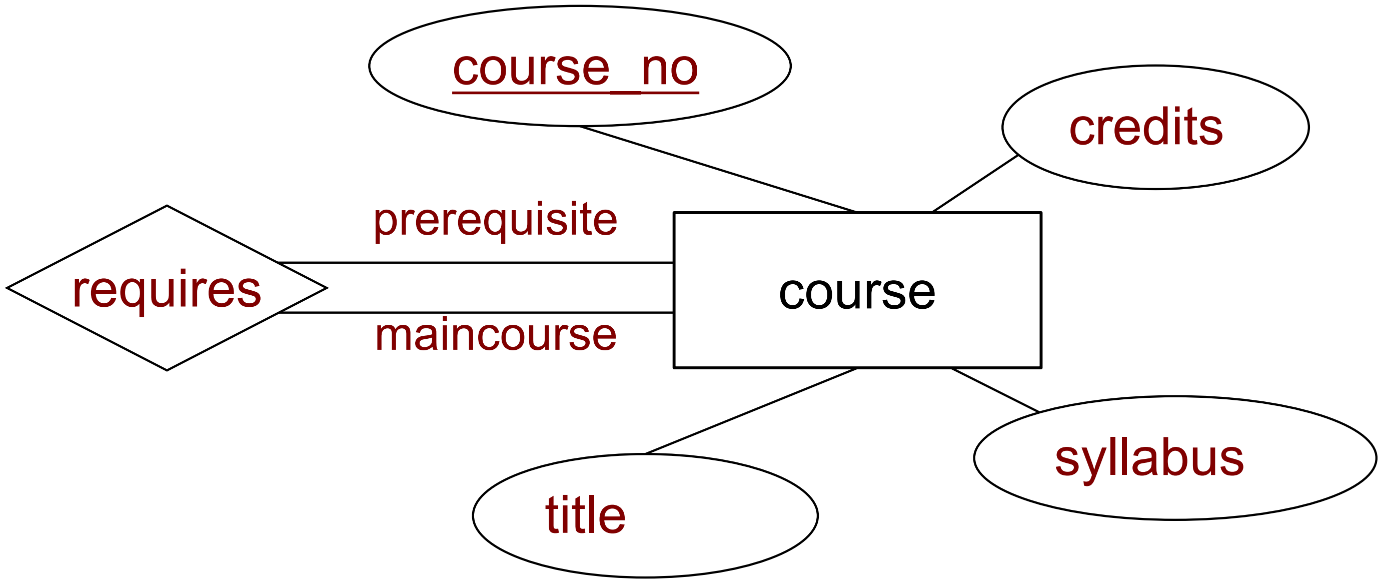
(a) course (b) course offering (c) student (d) instructor

Course

- “**course** including number, title, credits, syllabus, and prerequisites”

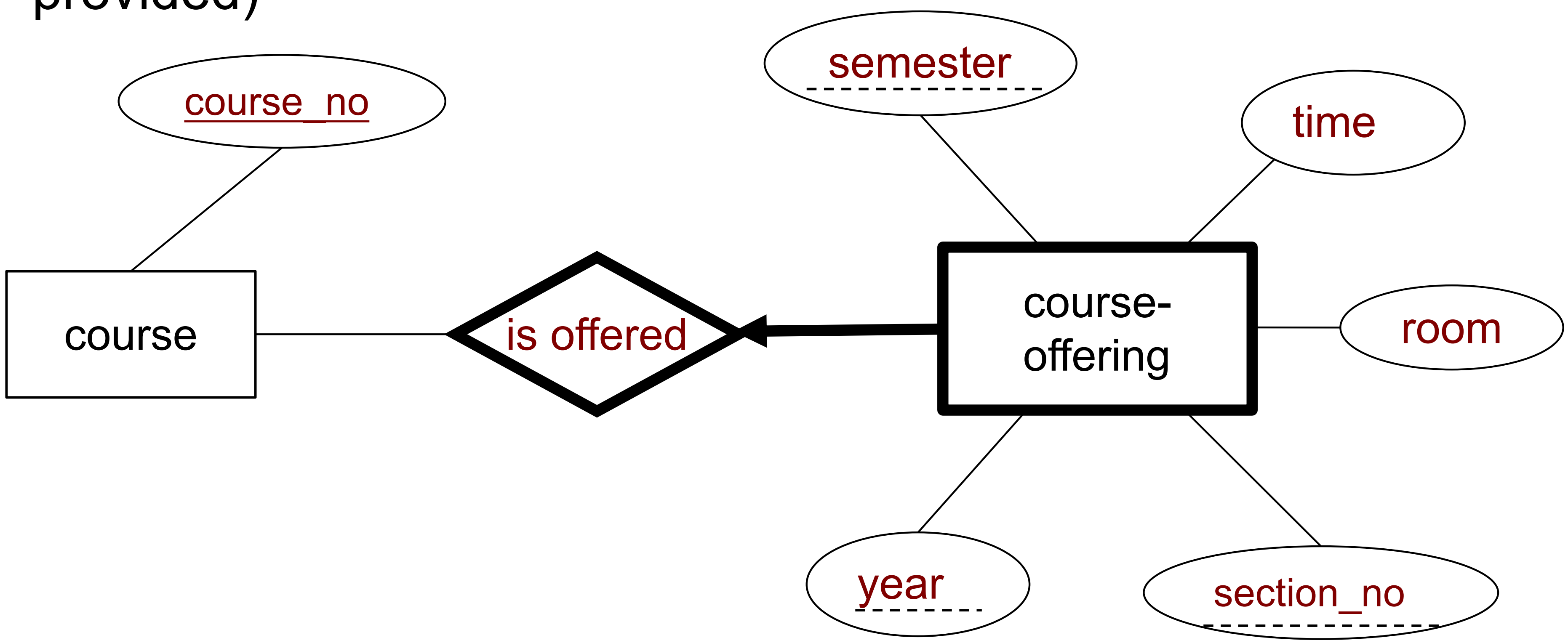
  Entity set? Attribute?

  Relationship set? Roles?



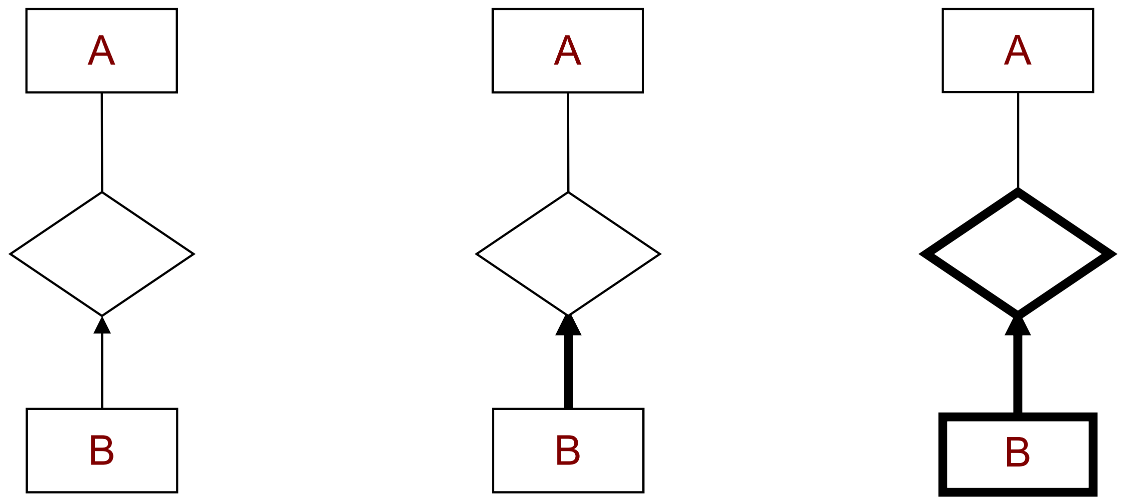
Course Offering

- "**course offering**, including course number, year, semester, section number, instructor(s), timings, and classroom"
- (course offering depends on the course that university provided)



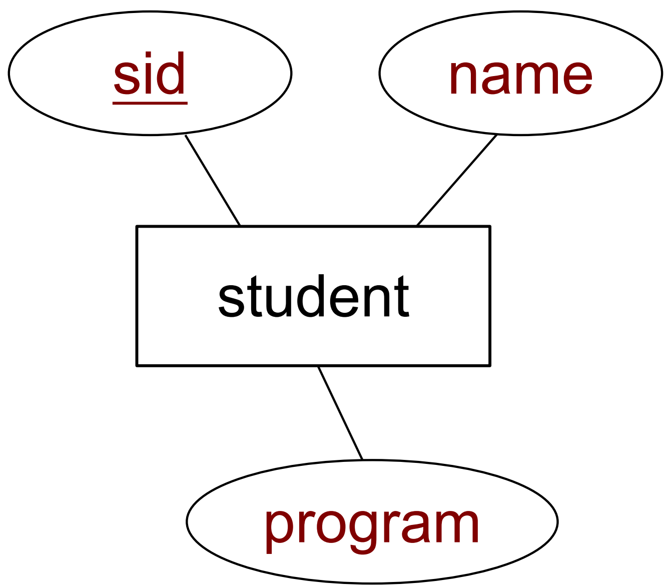
Weak Entity Set

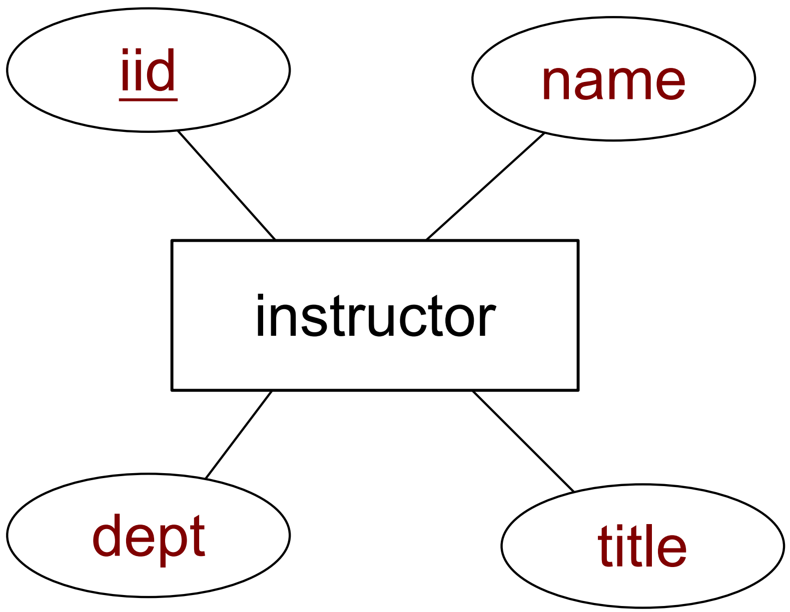
- A **weak entity set** can be identified uniquely only by considering the primary key of another (owner) entity set
  - Owner entity set and weak entity set must participate in **one-to-many** relationship set (one owner, many weak entities).
  - Weak entity set must have **total participation** in this identifying relationship set.



Student, Instructor

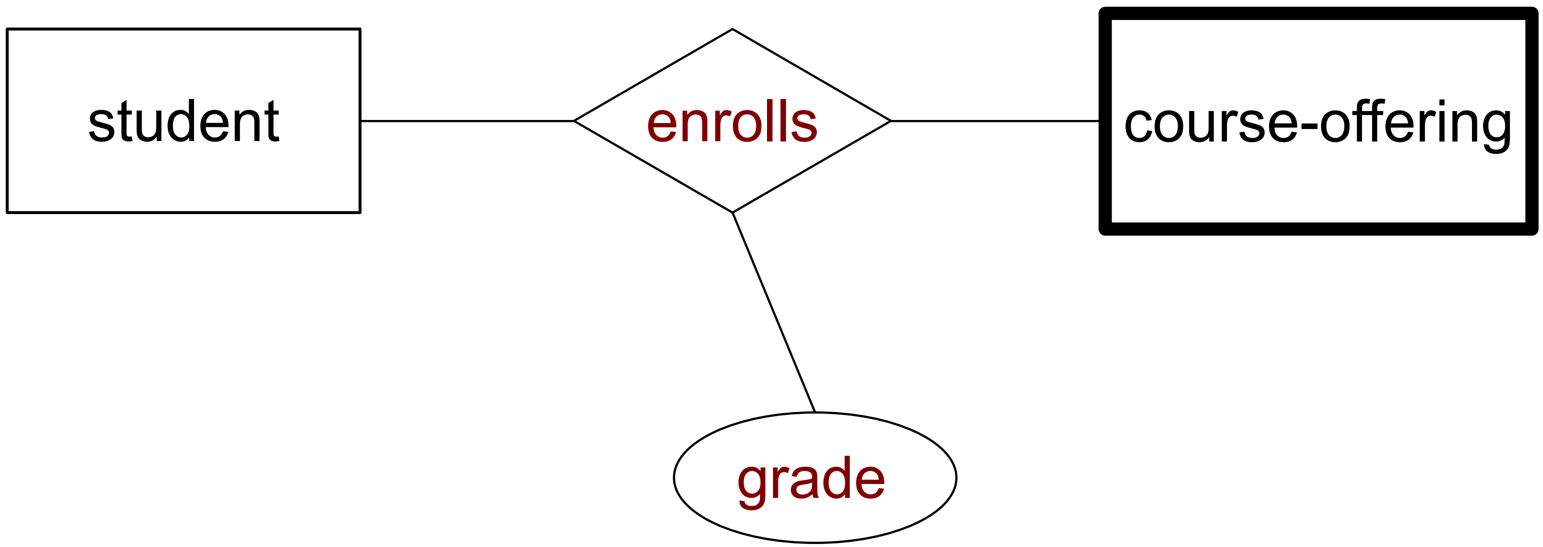
- "**student**, including student-id, name, and program"
- "**instructor**, including identification number, name, department, and title"





Enrollment

- "Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled."

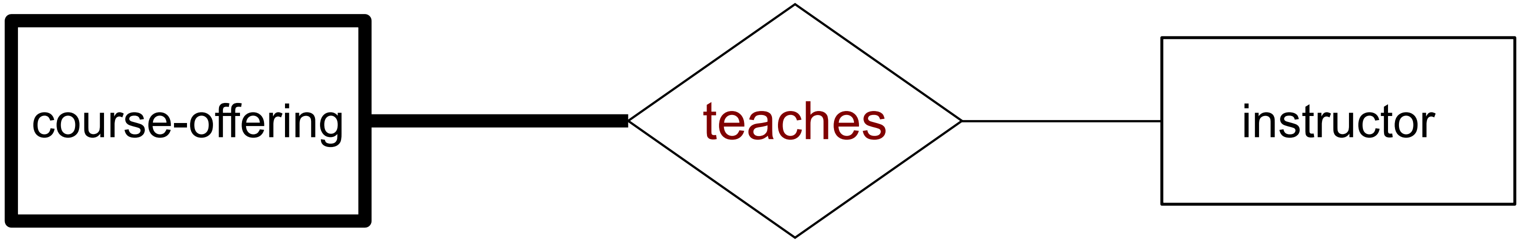


Requestment

- A university registrar's office maintains data about the following entities:
  a) **course**, including number, title, credits, syllabus, and prerequisites;
  b) **course offering**, including course number, year, semester, section number, instructor(s), timings, and classroom; (course offering depends on the course that university provided)
  c) **student**, including student-id, name, and program;
  d) **instructor**, including identification number, name, department, and title.
  - Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled.
- Construct an E-R diagram for the registrar's office. Document all assumptions that you make about the cardinality constraints and participation constraints.

Anymore ?

- Instructor teaches course…..



~~A course-offering does not have to have an instructor.~~

~~Partial participation~~

A course-offering has to have an instructor.

Total participation

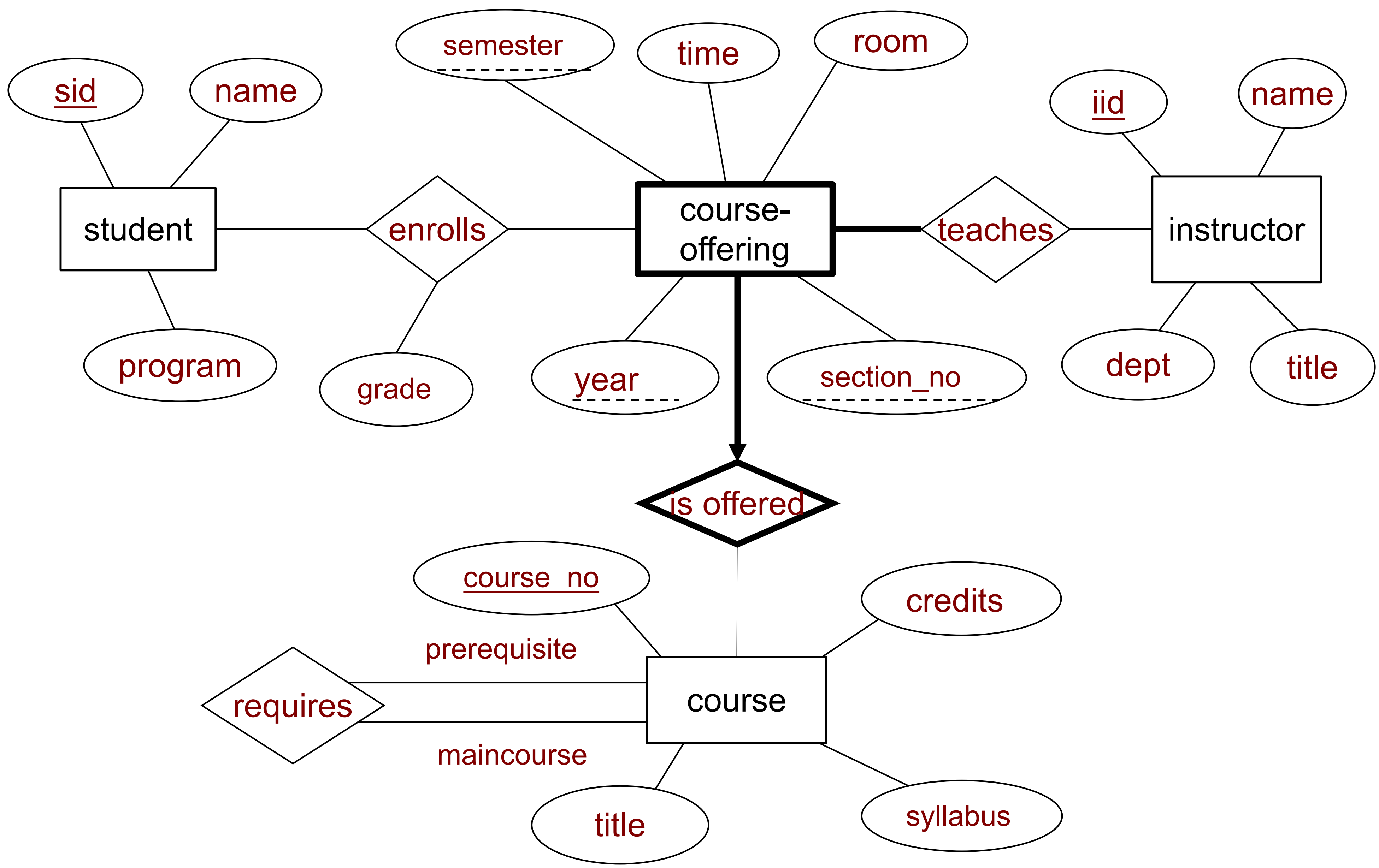


Figure 1 E-R diagram for a university registrar office.

More …

Based on **Figure 1**, modify the E-R Diagram so as to present the following information.

- Each instructor must belong to exactly one department.
- Each department includes dept-id, name, telephone number.
- Each department must have at least one instructor.

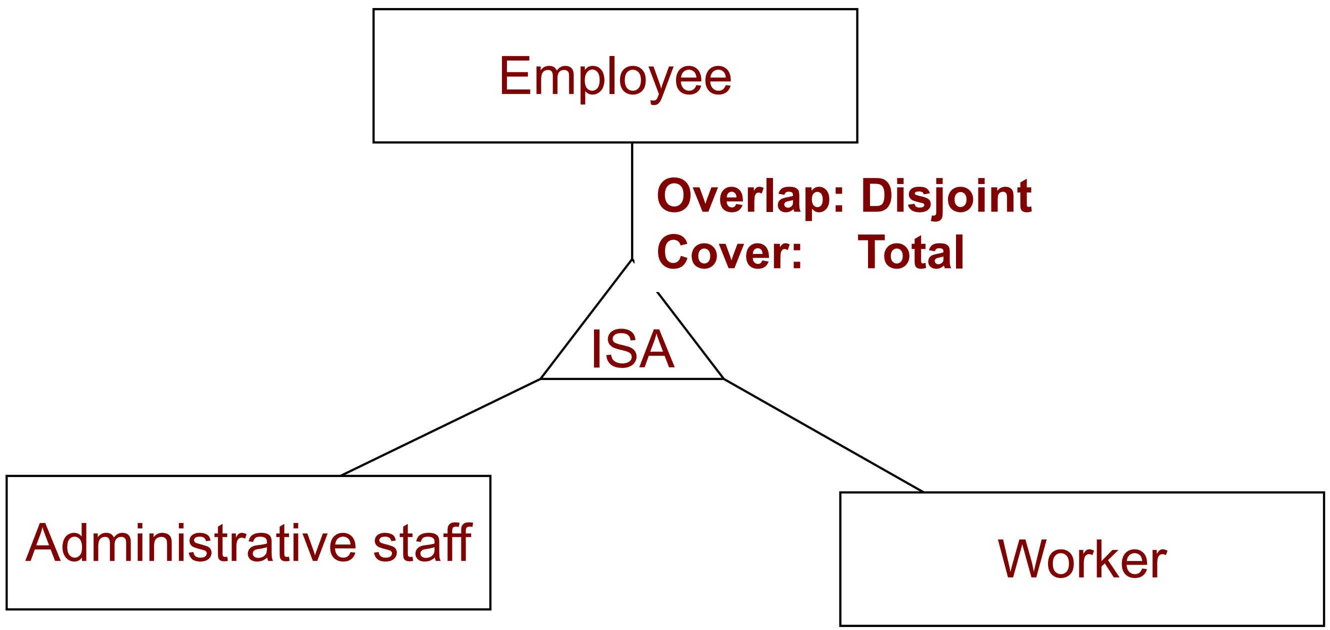
Exercise 2

- Design a database to organize the information about a factory and the products that are manufactured there. The relevant information is as follows:
  a) The factory has a number of **employees**. For each employee you need to store the name, employee number, and salary.
  b) Each employee must be an **administrative staff** or a **worker**, but not both.
  c) Administrative employees **must** take **seminars**. For each seminar we keep its id, name and date. For the administrative staff, you must store the grade received, for each seminar taken.
  d) The factory manufactures a number of **products** and each product is identified by a product id and has a name.
  e) Each worker is assigned to work on **exactly** one product; a product has multiple (**one or more**) workers assigned to it.
  f) A large number of **items** are manufactured for each product. Each item has a serial number and a color. Different items of the same product have different serial numbers. However, two items that belong to different products may have the same serial number.

Employee, Administrative Staff, Worker

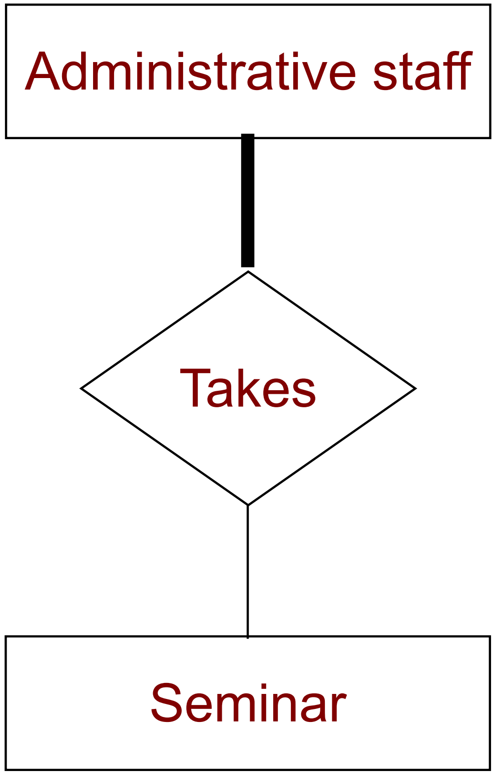
a) The factory has a number of **employees**. For each employee you need to store the name, employee number, and salary.

b) Each employee must be an **administrative staff** or a **worker**, but not both.



Seminar

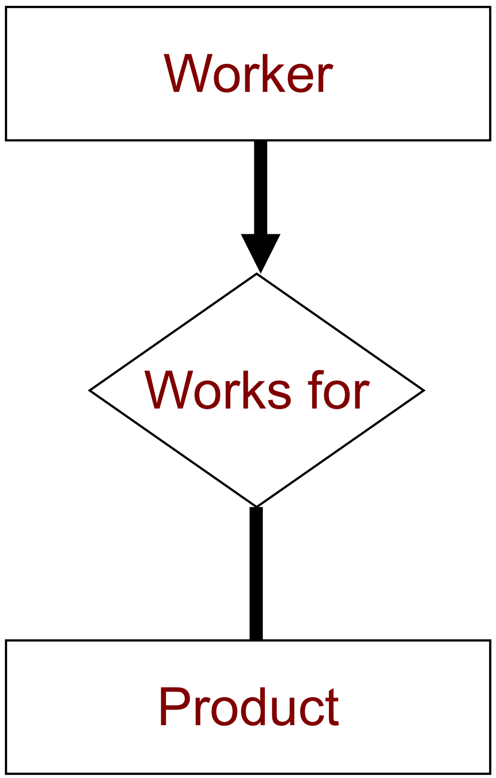
c) Administrative employees must take **seminars**. For each seminar we keep its id, name and date. For the administrative staff, you must store the grade received, for each seminar taken.



Product

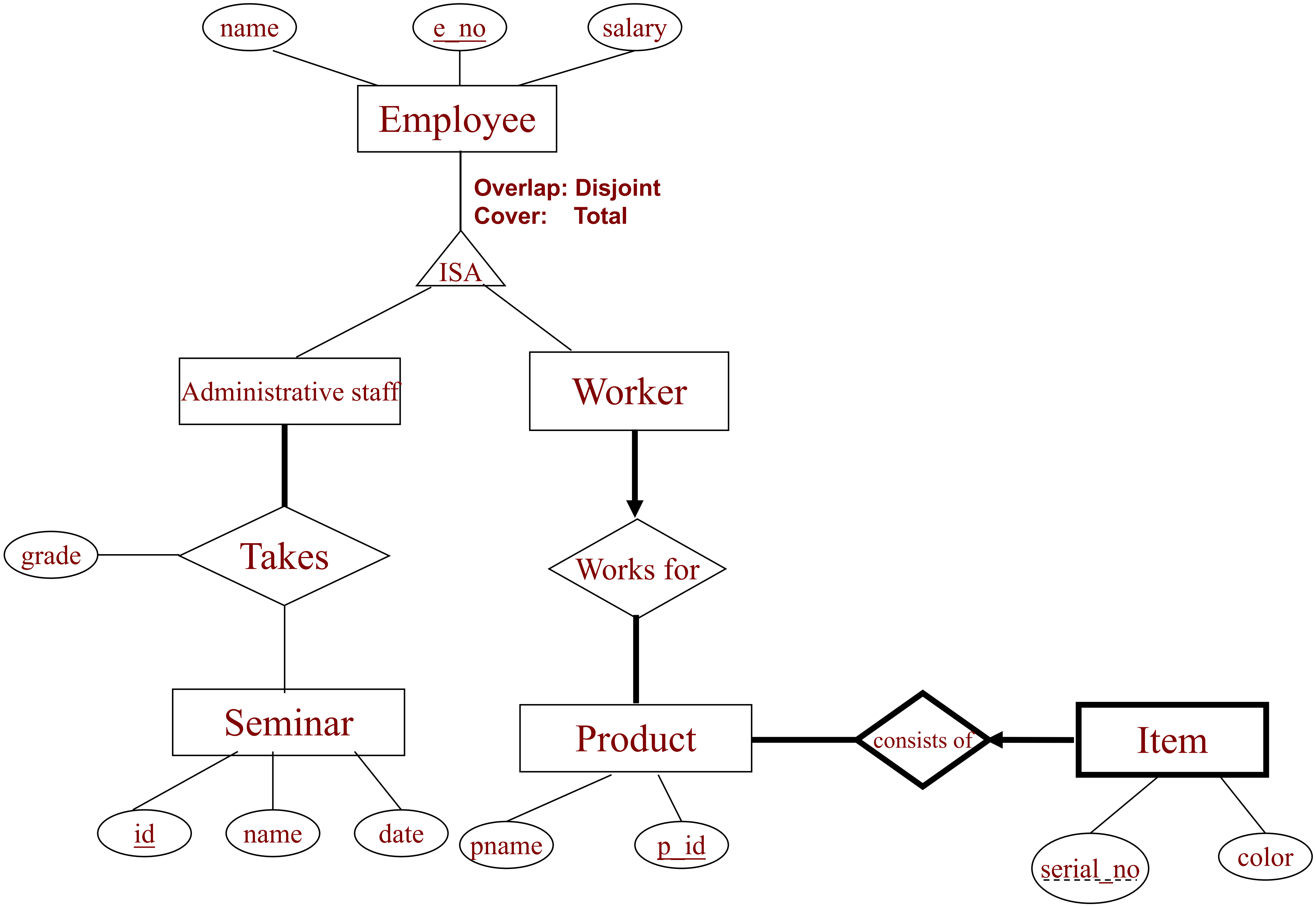
d) The factory manufactures a number of **products** and each product is identified by a product id and has a name.

e) Each worker is assigned to work on exactly one product; a product has multiple (one or more) workers assigned to it.



Item

f) A large number of **items** are manufactured for each product. Each item has a serial number and a color. Different items of the same product have different serial numbers. However, two items that belong to different products may have the same serial number.



name
e_no
salary
Employee
Overlap: Disjoint
Cover: Total
ISA
Administrative staff
Worker
grade
Takes
Works for
Seminar
Product
consists of
Item
id
name
date
pname
p_id
serial_no
color

Conclusion

- Notations of E-R Diagram
- Generally, we construct an E-R diagram, by identifying:
  - Entity sets
    - What should be entities?
    - Strong or weak entity set?
  - Relationship sets
    - Need to label roles?
  - Attributes
    - Can be inherited? (generalization or specification)
  - Participation constraints:
    - Total participation or partial participation?
  - Cardinality constraints
    - One to one, one to many or many to many?